

Tasbiha Asim

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PROFESSIONAL SUMMARY

Backend software engineer with 2+ years of production level Python experience, specializing in scalable backend services, API design, and cloud-native infrastructure. At Miro, owned end-to-end delivery of Python services deployed on Kubernetes (AWS EKS), with full CI/CD via GitHub Actions and Terraform. Holds an Erasmus Mundus MSc in Software Engineering and thrives in agile, cross-functional product teams.

WORK EXPERIENCE

Miro · Amsterdam, Netherlands

Software Engineer — AI Platform Team Aug 2025 – Aug 2026

- Implemented the integration that routes AI usage events (user prompts and model responses) to an internal governance service across three backend services (Python and Kotlin), enabling Microsoft Purview enterprise compliance reporting; handled gRPC client integration, Redis-based sequence tracking, context transfer, and error propagation.
- Implemented a new structured AI-interaction-logging producer on a backend AI service; delivered Input/Output and Model-Invocation event emission, gRPC channel wiring, session-ID-scoped request tracking, and ordered sequencing of tool and model invocations across multi-step agentic run cycles.
- Led migration of an image-processing path from PIL to OpenCV in a core Python backend service that made image decoding 2× faster. Shipped via feature flag, monitored decode-failure metrics, and added architectural tests to enforce the change.
- Introduced OpenTelemetry trace context into the AI run context for end-to-end tracing; maintained service-health dashboard panels via Prometheus.

Software Engineer — AI Platform Team Feb 2025 – Aug 2025

- Optimised the Screenshot Scanner Service (Python backend) by transmitting resource IDs instead of full image payloads, reducing inter-service latency.
- Designed a GA right-sizing strategy: profiled CPU/memory under varying load, tuned Kubernetes resource requests/limits, and authored alerting runbook to enable General Availability launch.
- Deprecated 3 legacy ML services, delivering ~\$128K in estimated annual infrastructure cost savings.

Software Engineer — Data Engineering Team Aug 2024 – Feb 2025

- Built the first working prototype of an Enterprise Guard board-title keyword detector service in PySpark.
- Migrated 5 SaaS data sources to Airbyte-backed dbt models; built and maintained dbt transformation models on Snowflake (PostgreSQL-compatible SQL).

Netherlands Organisation for Applied Scientific Research (TNO) · Eindhoven, Netherlands

Thesis Research Intern Dec 2023 – Aug 2024

- Designed and implemented a lane-change detection algorithm from scratch over the Waymo Open Dataset to extract overtaking and lane-change scenarios from 3D object tracks and road geometry features at scale.
- Conducted independent research on AV behaviour in mixed traffic, developing mathematical models of overtaking manoeuvres and investigating how human drivers respond to lane-changing events using the extracted scenarios.

EDUCATION

MSc Software Engineering — Erasmus Mundus Joint Masters Sep 2022 – Aug 2024

University of L'Aquila (IT) · LUT University (FI) · Vrije Universiteit Amsterdam (NL)

- Full-ride Erasmus Mundus merit-based scholarship.
- Coursework: Microservices Architecture, Self-Adaptive Systems, Green Software Engineering, Requirements Engineering, Software Testing & Quality.

BSc Computer Science Jul 2018 – Jun 2022

Habib University · Karachi, Pakistan

- HUTOPS 100% merit-based scholarship.
- Thesis: Knowledge-driven WhatsApp-based Financial Advisor (NLP + knowledge engineering).

TECHNICAL SKILLS

Languages	Python 3.x (primary), SQL, Kotlin, Go, JavaScript/TypeScript
Backend & APIs	FastAPI, REST APIs, gRPC, Protobuf, PySpark, SQLAlchemy
Async & Messaging	Celery, RabbitMQ, Redis
Databases	Snowflake (PostgreSQL-compatible), PostgreSQL, Redis
Cloud	AWS (EKS, Bedrock, DynamoDB), Kubernetes, Docker, Terraform, Helm
Observability	GitHub Actions, Prometheus, OpenTelemetry, integration testing
Data Engineering	dbt, Airbyte, Snowflake
Frontend	React, Angular (familiarity)